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Pet Door Insert for a French Door

Abstract

A panel insert for modifying a French door to make the French door pet accessible. The panel insert is positioned and secured within a frame of one of the French door panels. An upper opening and a lower opening are cut or formed into a body of the panel insert. A window component is secured within the upper opening and a pet door component is secured within the lower opening. The pet door component is configured to allow an animal to travel through the French door without human intervention. A pet accessible French door with an integrated panel insert is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/560,842, which was filed on Mar. 4, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention generally relates to a pet door. More specifically, the present invention relates to an apparatus for retrofitting a French door that is configured to enable an animal to enter and exit through the French door without human intervention. Accordingly, the present specification makes specific reference thereto. However, it is to be appreciated that aspects of the present invention are also equally amenable to other like applications, devices and methods of manufacture.

BACKGROUND

[0003] Many homes have dog or pet doors that allow domesticated pets to come and go as they please. However, a good number of homes have French doors, which do not currently accommodate a dog door. In these cases, pet owners have to leave the doors open so that their dog can enter and exit. This is not practical for many reasons. Leaving a dog outside in different weather conditions can be detrimental to the animal. Indoor accidents may occur more frequently if the dog cannot get outside to relieve themselves.

[0004] A pet door or pet flap is a small opening cut or formed into an exterior door designed to allow pets to enter and exit a building on their own without needing a human to open the door for the animal. Originally created as simple holes, the modern pet door is a typically a panel or flap that the pet can simply push through. These devices offer a degree of protection against wind, rain, and larger-bodied intruders entering the dwelling. Similar hatches can let dogs through fences or other solid barriers.

[0005] French doors are a type of door that originated in France and are known for their distinctive design, which features glass panels that extend for most of the door's length and are often divided by a grid. French doors are usually installed as a pair of hinged doors that open from the center, either swinging or sliding. Traditional French doors are wood framed, but can also be vinyl, aluminum, or fiberglass in construction as well. These doors are typically constructed as a pair of doors that open from the center, enhancing natural light and visual connectivity between spaces. Because of the significant glass surface area, French doors are unable to accommodate existing pet doors.

[0006] Therefore, there exists a long-felt need in the art for a pet or dog door that can retrofit an existing French door. There is also a long-felt need in the art for a way for owners of existing homes with French doors to accommodate their pets' needs. Further, there is a long-felt need in the art for a way for pets to reenter the home when needed to escape extreme weather conditions. Finally, there is a long-felt need in the art for an alternative for the need for pet owners to leave their exterior door open to accommodate their pet.

[0007] In this manner, the improved pet door insert of the present invention accomplishes all of the foregoing objectives, thereby providing an easy solution for ensuring that homes with French doors are able to accommodate their pets' needs by creating a safe and convenient pet door. A primary feature of the present invention is an insert for a French door that creates a pet door to enable animals to exit. Advantageously, the French door insert allows pets to exit through the French door to relieve themselves outside and to reenter the home as desired without human intervention. Finally, the improved French door insert of the present invention is capable of eliminating the need for pet owners to leave their exterior door open to accommodate their dog.

SUMMARY

[0008] The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to

present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0009] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a panel insert for a French door. The French door comprises a first door panel and a second door panel. The first and second door panels each comprise a top rail, a bottom rail, a hinge style, and a lock style. The top and bottom rails and the hinge and lock styles form a frame for a glass portion of each door panel.

[0010] The panel insert of the present invention comprises a body comprising a perimeter and a lower opening. The body is sized to fit within an opening in one of the door panels of the French door created by removing the respective glass portion between the top and bottom rails and the hinge and lock styles of the respective door panel. The body may further comprise a transition element surrounding the perimeter of the body. The body may also further comprise an upper opening.

[0011] The panel insert may further comprise a window component. The window component is configured to fit within the upper opening of the body. The window component may be fixed or movable within the upper opening. The window component may comprise a frame and a window. The window component may further comprise a window locking element for securing the window. The window component may further comprise a screen. The window component may further comprise a window cover. The window cover is configured to completely cover the upper opening.

[0012] The panel insert may further comprise a pet door component. The pet door component is configured to fit within the lower opening of the body. The pet door component comprises a movable portion pivotally mounted within the lower opening. The movable portion is configured to swing both inward and outward. The pet door component further comprises a movable portion locking element for securing the movable portion. The pet door component may further comprise a pet door cover. The pet door cover is configured to completely cover and secure the lower opening.

[0013] The subject matter disclosed and claimed herein, in another embodiment thereof, comprises a pet accessible French door. The pet accessible French door comprises a first door panel and a second door panel. The first and second door panels each comprise a top rail, a bottom rail, a hinge style, and a lock style. The top and bottom rails and the hinge and lock styles form a frame for a glass portion of each door panel.

[0014] The pet accessible French door further comprises a panel insert. The panel insert comprises a body comprising a perimeter and a lower opening. The body is sized to fit within an opening in one of the first or second door panels created by removing the respective glass portion between the top and bottom rails and the hinge and lock styles of the respective door panel. The body may further comprise a transition element surrounding the perimeter of the body. The body may further comprise an upper opening.

[0015] The panel insert may further comprise a window component. The window component is configured to fit within the upper opening of the body. The window component may be fixed or movable within the upper opening. The window component may comprise a frame and a window. The window component may further comprise a window locking element for securing the window. The window component may further comprise a screen. The window component may further comprise a window cover. The window cover is configured to completely cover the upper opening.

[0016] The panel insert may further comprise a pet door component. The pet door component is configured to fit within the lower opening of the body. The pet door component comprises a movable portion pivotally mounted within the lower opening. The movable portion is configured to swing both inward and outward. The pet door component further comprises a movable portion locking element for securing the movable portion. The pet door component may further comprise a pet door cover. The pet door cover is configured to completely cover and secure the lower opening.

[0017] To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the

annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and is intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

[0019] FIG. 1 illustrates a front view of a French door for use with a panel insert of one embodiment of the present invention in accordance with the disclosed architecture.

[0020] FIG. 2 illustrates a front view of one embodiment of a panel insert of the present invention for use with a French door in accordance with the disclosed architecture.

[0021] FIG. 3 illustrates a front view of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0022] FIG. 4 illustrates a front view of a window component of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0023] FIG. 5 illustrates a side view of the window component of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0024] FIG. 6 illustrates a perspective view of a pet door component of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0025] FIG. 7 illustrates a perspective view of the pet door component of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0026] FIG. 8 illustrates a front view of the pet door component of the panel insert of the present invention for use with the French door in accordance with the disclosed architecture.

[0027] FIG. 9 illustrates a perspective view of one embodiment of a pet accessible French door of the present invention in accordance with the disclosed architecture.

DETAILED DESCRIPTION

[0028] The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They do not intend as an exhaustive description of the invention or do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

[0029] The present invention, in one exemplary embodiment, is a specially designed pet door insert for use with French doors that enables a dog to enter and exit the home while keeping the door closed. The invention is comprised of an insert that goes in a French door to replace the glass portion of the door. In one embodiment, the insert may measure approximately 20"×64"×¾". The insert may be constructed from plywood with no knots. The insert may comprise an opening cut out for a top glass portion of the door, which may be approximately 18"×36". An optional window may be added that measures approximately 17.5"×35.5". The insert also includes a flap that enables the animal to enter and exit with ease. A door is included to cover the flap when needed.

[0030] Referring initially to the drawings, FIGS. 1-8 illustrates a panel insert 100 for use with a

French door **10** to make the French door **10** pet accessible. As illustrated in FIG. 1, the French door **10** comprises a first door panel **20** and a second door panel **30**. The first and second door panels **20** and **30** each comprise a top rail **32**, a bottom rail **34**, a hinge style **36**, and a lock style **38**. The top and bottom rails **32** and **34** and the hinge and lock styles **36** and **38** form a frame or an opening **50** for a glass portion **40** of each door panel **20** and **30**. The French door **10** may be of any French door configuration that has a pair of hinged doors that open from the center, either swinging or sliding. [0031] As illustrated in FIG. 2, the panel insert **100** comprises a body **110**. The body is typically constructed from wood but may be constructed from any material used in the art of door or window construction. The body **110** is typically rectangular and comprises a perimeter **112** and a lower opening **118**. The body **110** is sized to fit within the opening **50** in one of the door panels **20** or **30** of the French door **10** that created by removing the respective glass portion **40** between the top and bottom rails **32** and **34** and the hinge and lock styles **36** and **38** of the respective door panel **20** or **30**. The lower opening **118** is positioned in a portion of a lower half of the body **118** and is dimensioned as discussed infra.

[0032] As illustrated in FIG. 3, the body **110** may further comprise a transition element **114**. The transition element **114** surrounds the perimeter **112** of the body **110**. The transition element **114** may be a metal, rubber, plastic, or foam weatherstripping and may also be designed to adjustably seal or attach the body **110** within the opening **50** of the respective door panel **20** or **30**. The body **110** may also further comprise an upper opening **116**. The upper opening **116** is positioned in a portion of an upper half of the body **118** and is dimensioned as discussed infra.

[0033] As illustrated in FIGS. 4 and 5, the panel insert **100** may further comprise a window component **120**. The window component **120** is dimensioned to fit within the upper opening **116** in the body **110** and may be any size as desired. The window component **120** comprises a window **124**. The window **124** may comprise a window frame **122** for securing the window **124** within the upper opening **116**. The window **122** may be unopenable, such as a glass panel directly integrated into the body **110** within the upper opening **116**, or at least partially openable. The window **122** may be any type, size, shape, and configuration of window such as but not limited to a casement window, an awning window, a slider window, a single hung window, a double hung window, a bay window, a bow window, a garden window, a picture window, a transom window, a stationary window, a hopper window, a jalousie window, a tilt and turn window, a pivot window, an arched window, etc.

[0034] The window component **120** may further comprise a window lock **126**. The window lock **126** is typically a latch, clasp, magnetic lock, or similar window locking element designed to lock the window **122** when closed. The window component **120** may further comprise a screen **128**. The screen **128** may be used alone within the upper opening **116**, or in conjunction with the window **122**. The window component **120** may further comprise a window cover **130**. The window cover **130** is a panel designed to close over, or slide in place behind, the window **122** on an inside side of the French door **10**. The window cover **130** is configured to further secure and block access through the upper opening **116**.

[0035] As illustrated in FIGS. 6-8, the panel insert **100** further comprises a pet door component **140**. The pet door component **140** is dimensioned to fit within the lower opening **118** in the body **110** and may be any size as desired. The pet door component **140** comprises a movable portion **144**. The movable portion **144** is pivotally mounted within the lower opening **118** and configured to swing inward and outward. The pet door component **140** may further comprise a pet door frame **142** for securing the movable portion **144** within the lower opening **118**.

[0036] The pet door component **140** may be any size, shape, and configuration of a pet door, such as, but not limited to, a manual flap door, a magnetic door, a microchip door, an infrared door, a sliding glass insert door, an electronic door, a screen door, a telescoping door, etc. In one embodiment, the movable portion **144** is a flap. The flap **144** ensures air cannot travel through the lower opening **118** of the body **110**. In one embodiment, the flap **144** may be configured to

magnetic magnetically engage the lower opening **118**.

[0037] The pet door component **140** may further comprise a hinge **148**. The hinge **148** pivotally mounts the movable portion or flap **144** to the body **110** within the lower opening **118** and permits passage of an animal in both directions as the flap **144** can swing inward and outward. The hinge **148** may be a top mount hinge securing the movable portion **144** from the top, or a side mounted hinge securing the movable portion **144** from the side as illustrated in FIG. 7.

[0038] The pet door component **140** may further comprise a pet door lock **146**. The pet door lock **146** is typically a latch, clasp, or similar locking element designed to lock the movable portion **144** when closed. The pet door component **140** may further comprise a pet door cover **150**. The pet door cover **150** is a panel designed to close over, or slide in place behind the movable portion **144** on an inside side of the French door **10**. The pet door cover **150** is configured to further secure and block access through the lower opening **118** and may further be securable with a cover locking element **152**. As illustrated in FIG. 8, the movable portion **144** may be adjusted in size within the lower opening **118** to accommodate different sized pets.

[0039] In an additional embodiment, FIG. 9 illustrates a pet accessible French door **200**. The pet accessible French door **200** comprises a first door panel **202**, a second door panel **204**, and a panel insert **100**. The first and second door panels **202** and **204** each comprise a top rail **32**, a bottom rail **34**, a hinge style **36**, and a lock style **38**. The top and bottom rails **32** and **34** and the hinge and lock styles **36** and **38** form a frame or an opening **50** for a glass portion **40** of each door panel **202** and **204**.

[0040] The panel insert **100** comprises a body **110**. The body is typically constructed from wood, but may be constructed from any material used in the art of door or window construction. The body **110** is typically rectangular and comprises a perimeter **112** and a lower opening **118**. The body **110** is sized to fit within the opening **50** in one of the door panels **202** or **204** of the pet accessible French door **200** that created by removing the respective glass portion **40** between the top and bottom rails **32** and **34** and the hinge and lock styles **36** and **38** of the respective door panel **20** or **30**. The lower opening **118** is positioned in a portion of a lower half of the body **118** and is dimensioned as discussed infra.

[0041] The body **110** may further comprise a transition element **114**. The transition element **114** surrounds the perimeter **112** of the body **110**. The transition element **114** may be a metal, rubber, plastic, or foam weatherstripping and may also be designed to adjustably seal or attach the body **110** within the opening **50** of the respective door panel **20** or **30**. The body **110** may also further comprise an upper opening **116**. The upper opening **116** is positioned in a portion of an upper half of the body **118** and is dimensioned as discussed infra.

[0042] The panel insert **100** may further comprise a window component **120**. The window component **120** is dimensioned to fit within the upper opening **116** in the body **110** and may be any size as desired. The window component **120** comprises a window **124**. The window **124** may comprise a window frame **122** for securing the window **124** within the upper opening **116**. The window **122** may be unopenable, such as a glass panel directly integrated into the body **110** within the upper opening **116**, or at least partially openable. The window **122** may be any type, size, shape, and configuration of window such as but not limited to a casement window, an awning window, a slider window, a single hung window, a double hung window, a bay window, a bow window, a garden window, a picture window, a transom window, a stationary window, a hopper window, a jalousie window, a tilt and turn window, a pivot window, an arched window, etc.

[0043] The window component **120** may further comprise a window lock **126**. The window lock **126** is typically a latch, clasp, or similar window locking element designed to lock the window **122** when closed. The window component **120** may further comprise a screen **128**. The screen **128** may be used alone within the upper opening **116**, or in conjunction with the window **122**. The window component **120** may further comprise a window cover **130**. The window cover **130** is a panel designed to close over, or slide in place behind the window **122** on an inside side of the pet

accessible French door **200**. The window cover **130** is configured to further secure and block access through the upper opening **116**.

[0044] The panel insert **100** further comprises a pet door component **140**. The pet door component **140** is dimensioned to fit within the lower opening **118** in the body **110** and may be any size as desired. The pet door component **140** comprises a movable portion **144**. The movable portion **144** is pivotally mounted within the lower opening **118** and configured to swing inward and outward. The pet door component **140** may further comprise a pet door frame **142** for securing the movable portion **144** within the lower opening **118**.

[0045] The pet door component **140** may be any size, shape, and configuration of a pet door, such as, but not limited to, a manual flap door, a magnetic door, a microchip door, an infrared door, a sliding glass insert door, an electronic door, a screen door, a telescoping door, etc. In one embodiment, the movable portion **144** is a flap. The flap **144** ensures air cannot travel through the lower opening **118** of the body **110**. In one embodiment, the flap **144** may be configured to magnetically engage the lower opening **118**.

[0046] The pet door component **140** may further comprise a hinge **148**. The hinge **148** pivotally mounts the movable portion or flap **144** to the body **110** within the lower opening **118** and permits passage of an animal in both directions as the flap **144** can swing inward and outward. The hinge **148** may be a top mount hinge securing the movable portion **144** from the top, or a side mounted hinge securing the movable portion **144** from the side.

[0047] The pet door component **140** may further comprise a pet door lock **146**. The pet door lock **146** is typically a latch, clasp, or similar locking element designed to lock the movable portion **144** when closed. The pet door component **140** may further comprise a pet door cover **150**. The pet door cover **150** is a panel designed to close over, or slide in place behind the movable portion **144** on an inside side of the pet accessible French door **200**. The pet door cover **150** is configured to further secure and block access through the lower opening **118** and may further be securable with a cover locking element **152**.

[0048] It is contemplated that the panel insert **100** and pet accessible French door **200** constructed in accordance with the present invention will be tailored and adjusted by those of ordinary skill in the art to accommodate various levels of performance demand imparted during actual use. Accordingly, while this invention has been described by reference to certain specific embodiments and examples, it will be understood that this invention is capable of further modifications. This application is, therefore, intended to cover any variations, uses or adaptations of the invention following the general principles thereof, and including such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and fall within the limits of the appended claims.

[0049] Notwithstanding the foregoing, the panel insert **100** and pet accessible French door **200** of the present invention and its various structural components can be of any suitable size, shape, and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the shape and size of the panel insert **100** and pet accessible French door **200** and its various components and material, as shown in the FIGS. are for illustrative purposes only, and that many other shapes and sizes of the panel insert **100** and pet accessible French door **200** are well within the scope of the present disclosure. Although the dimensions of the panel insert **100** and pet accessible French door **200** are important design parameters, the panel insert **100** and pet accessible French door **200** and its components may be of any shape or size that ensures optimal performance during use and/or that suits user need and/or preference.

[0050] What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are

possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. A panel insert for a French door, the panel insert comprising: a body comprising a perimeter and a lower opening; and a pet door component configured to fit within the lower opening; and wherein the pet door component comprises a movable portion pivotally mounted within the lower opening and configured to swing inward and outward; and wherein the body is configured to fit within an opening in a door panel of the French door.
2. The panel insert of claim 1, wherein the body further comprises a transition element surrounding the perimeter for adjustably fitting the body within the opening in the door panel of the French door.
3. The panel insert of claim 1, wherein the pet door component further comprises a hinge for pivotally mounting the movable portion to the body.
4. The panel insert of claim 1, wherein the movable portion is a flap.
5. The panel insert of claim 1, wherein the pet door component further comprises a lower opening cover configured to block access through the lower opening.
6. A panel insert for a French door, the panel insert comprising: a body comprising a perimeter, an upper opening, and a lower opening; a window component configured to fit within the upper opening; and a pet door component configured to fit within the lower opening; and wherein the pet door component comprises a movable portion pivotally mounted within the lower opening and configured to swing inward and outward; and wherein the body is configured to fit within an opening in a door panel of the French door.
7. The panel insert of claim 6, wherein the pet door component further comprises a lower opening cover configured to block access through the lower opening.
8. The panel insert of claim 6, wherein the window component is a glass panel integrated into the body within the upper opening.
9. The panel insert of claim 6, wherein the window component comprises a window configured to partially open.
10. The panel insert of claim 6, wherein the window component comprises a screen.
11. The panel insert of claim 6, wherein the window component comprises a locking element.
12. The panel insert of claim 6, wherein the window component comprises a window configured to partially open and an upper opening cover configured to block access through the upper opening.
13. The panel insert of claim 6, wherein the window component comprises a window configured to partially open and the movable portion of the pet door component is a flap.
14. The panel insert of claim 6, wherein the pet door component further comprises a locking element.
15. A pet accessible French door comprising: a French door component comprising at least one door panel comprising a top rail, a bottom rail, a hinge style, and a lock style; and a panel insert configured to fit within an opening defined by the top and bottom rails and the hinge and lock styles of the French door, the panel insert comprising: a body comprising a perimeter, an upper opening, and a lower opening; a window component configured to fit within the upper opening; and a pet door component configured to fit within the lower opening; and wherein the pet door component comprises a movable portion pivotally mounted within the lower opening and configured to swing inward and outward.
16. The pet accessible French door of claim 15, wherein the movable portion of the pet door

- component further comprises a top hinge for pivotally mounting the movable portion to the body.
- 17.** The pet accessible French door of claim 15, wherein the movable portion of the pet door component further comprises a side hinge for pivotally mounting the movable portion to the body.
- 18.** The pet accessible French door of claim 15, wherein the window component comprises a window configured to partially open and a locking element for securing the window.
- 19.** The pet accessible French door of claim 15, wherein the pet door component further comprises a lower opening cover configured to block access through the lower opening.
- 20.** The pet accessible French door of claim 15, wherein the window component comprises a window configured to partially open, a screen, and an upper opening cover configured to block access through the upper opening.
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